4.3 Biodiversity

Understanding Genetic, Species, and Ecological Diversity in Marine Environments

Learning Objectives

01

Three Levels of Biodiversity

Explain biodiversity at genetic, species, and ecological levels

02

Key Terminology

Define species abundance, species richness, and allele frequency

03

Ecosystem Stability

Understand why high biodiversity leads to ecosystem resilience

04

Marine Services

Describe five critical services marine biodiversity provides

05

Environment Examples

Give named examples of stable, unstable, extreme, and non-extreme marine environments

What is Biodiversity?

> The degree of variation of organisms and ecosystems on Earth

Biodiversity encompasses all living things and the complex ecosystems they form. It's measured at three distinct but interconnected levels, each revealing different aspects of life's variety.

Species Diversity

Variety of different species in a habitat

Genetic Diversity

Variation of genes within species populations

Ecological Diversity

Range of ecosystems across regions

Key Concept: High biodiversity creates more stable, resilient ecosystems

Species Diversity

Species diversity measures the **variety** and **quantity** of organisms in an area, combining two essential components to define an ecosystem's richness.

Species Abundance

The number of individuals per species. High abundance means many individuals of each species.

Species Richness

The total number of different species. High richness indicates many different types of organisms.

Example: Coral reefs show high richness (many species) and high abundance (many individuals per species).

Genetic Diversity

Genetic diversity refers to the variety of **allele forms** within a species. Genes, made of DNA or RNA, carry instructions for traits, and different versions of these genes are called alleles.

1

Each species has a characteristic set of genes

2

Individuals possess unique combinations of alleles

3

Greater genetic variation enables better adaptation

4

Low diversity threatens population survival

Think of alleles like different editions of the same book—same story, slightly different details

Genetic Diversity in Populations

A **population** consists of all individuals of one species living in the same area at the same time. Each population has its own unique genetic makeup.

1

Population Genetics

Different populations have varying allele frequencies

2

Measuring Diversity

Calculate how common each allele is per gene

3

Adaptation Potential

Low genetic diversity reduces ability to adapt to environmental changes

Example: Koi fish display remarkable color pattern variation due to high genetic diversity, with each individual expressing different allele combinations.

Ecological Diversity

Ecological diversity measures the variation of ecosystems and habitats across regional or global scales at one time. An **ecosystem** includes all organisms (biota) and their interactions with the non-living environment.

Coral Reefs

Tropical shallow-water ecosystems

Seagrass Meadows

Coastal marine plant communities

Rocky Shores

Wave-exposed coastal habitats

Hydrothermal Vents

Deep-sea volcanic ecosystems

Challenge: Measuring ecological diversity is difficult because Earth's ecosystems gradually blend into one another rather than having distinct boundaries.

Three Levels Summary

Species	Number and abundance of species in an area	500 fish species on a coral reef
Genetic	Variety of alleles within a species	Different color patterns in koi fish
Ecological	Variety of ecosystems regionally or globally	Coral reefs, kelp forests, hydrothermal vents

Importance of Marine Biodiversity

Marine biodiversity provides essential services that sustain both ocean ecosystems and human civilization. These interconnected services demonstrate why protecting ocean life matters.

Ecosystem Stability

High diversity creates resilient systems resistant to environmental changes

Physical Protection

Coastal ecosystems shield shorelines from storms and erosion

Climate Regulation

Marine organisms absorb carbon dioxide and produce oxygen

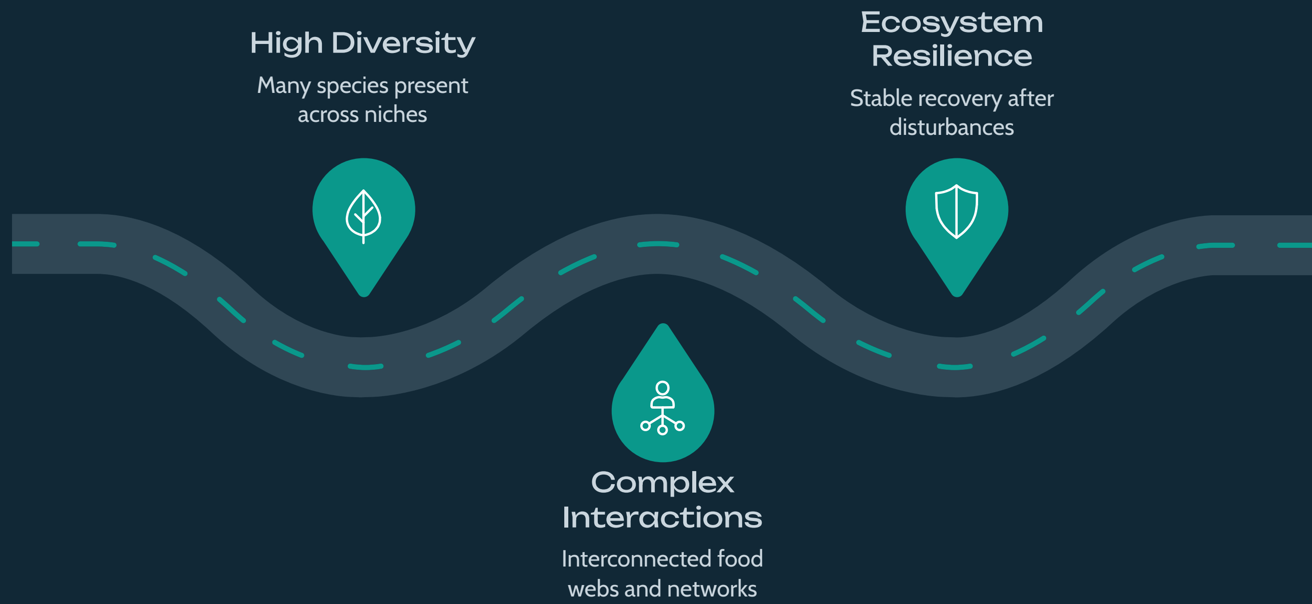
Food Resources

Oceans provide protein for billions of people worldwide

Medical Discoveries

Marine organisms yield compounds for treating diseases

Service 1: Maintaining Stable Ecosystems



The Biodiversity-Stability Connection

High species diversity is crucial for stable and resilient ecosystems, enabling them to better resist ecological disturbances due to:

- Complex interactions within diverse communities.
- Robust food webs with varied producers.
- Functional redundancy, ensuring critical roles persist even if some species decline.

Ecosystem Stability Factors

Two key factors determine an ecosystem's biodiversity:

1	2
Stability Constant environmental conditions allow species to adapt and thrive.	**Extremity** Abiotic factors (pH, temp, light, pressure) within tolerance zones support more species.

Stable + Non-extreme	**HIGH** biodiversity
Unstable OR Extreme	**LOW** biodiversity

Example: Coral Reef

STABLE & NON-EXTREME

Optimal Conditions

Abiotic factors near ideal for photosynthetic producers: warm stable temperatures, clear sunlight, stable pH

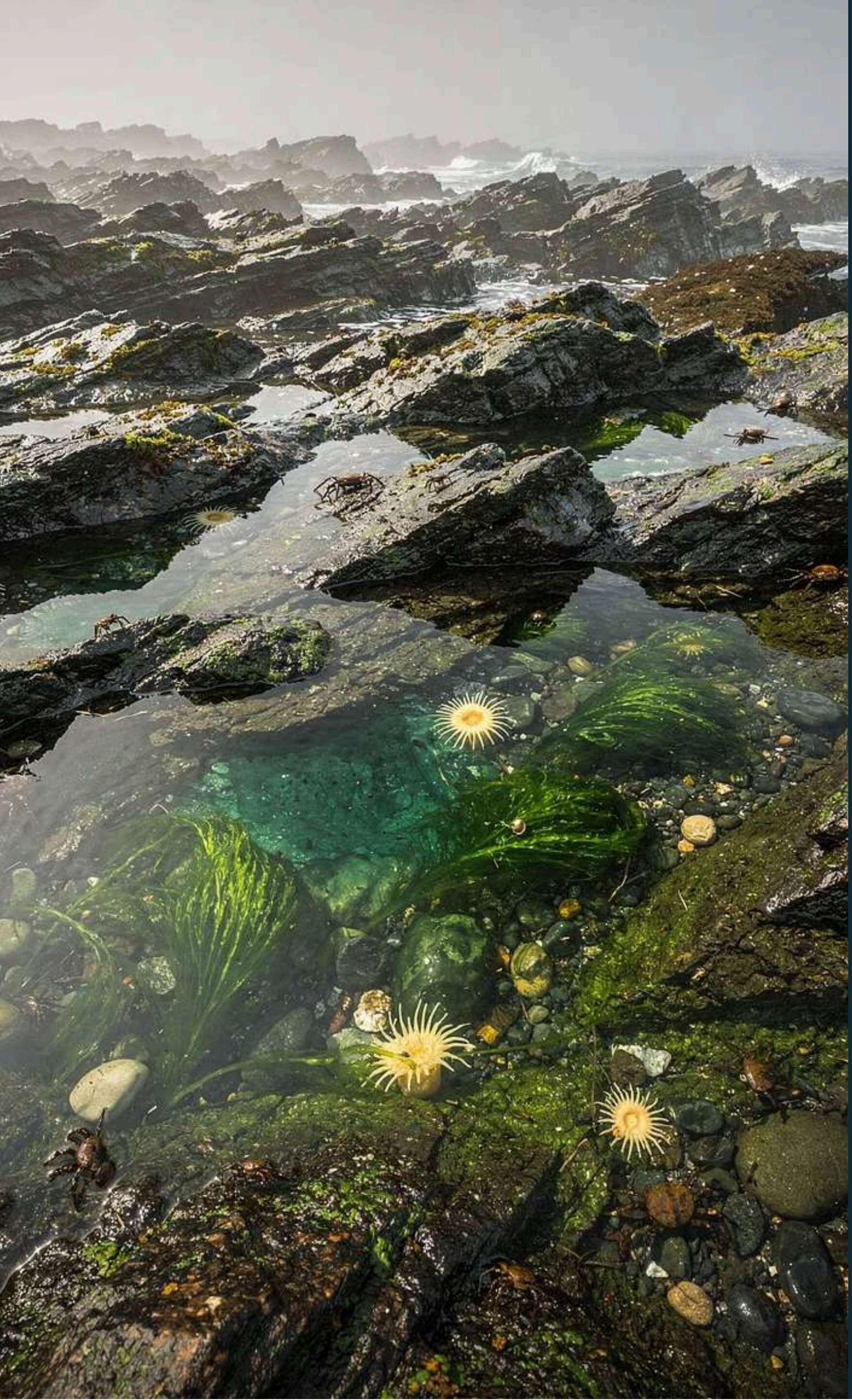
Environmental Consistency

Conditions remain predictable with minimal fluctuation throughout the year

Ecosystem Structure

Vibrant producer community forms foundation for long, complex food chains and diverse food webs

Result: HIGH BIODIVERSITY – Coral reefs contain over 25% of all marine species despite covering less than 1% of the ocean floor.



Example: Rocky Shore

STABLE & NON-EXTREME

Rocky shores demonstrate how physical structure enhances biodiversity even in dynamic coastal zones exposed to waves and tides.

Attachment Surface

Solid rock provides secure substrate for mollusks, barnacles, and seaweeds to anchor themselves

Protective Habitats

Tide pools and rock crevices offer refuge from predators, waves, and desiccation during low tide

Water Retention

Non-porous rock retains moisture better than sand, reducing drying stress for organisms

Result: Rocky shores support greater biodiversity than sandy shores due to superior habitat complexity and stability.

Example: Sandy Shore

UNSTABLE & NON-EXTREME

Sandy shores experience constant physical disturbance despite having temperatures and pH levels within normal ranges.

Challenges for Organisms

- Constant exposure to wave impact and erosion
- No solid attachment surface for anchoring
- Sandy substrate shifts and washes away easily
- pH and temperature near optimal (non-extreme)
- But continuous disturbance creates instability

Result: LOW BIODIVERSITY – Only mobile, burrowing species like worms, clams, and crabs can survive.

Example: Hydrothermal Vent

UNSTABLE & EXTREME

Hydrothermal vents are Earth's most extreme environments, pushing the limits of where life can exist.

Extreme Temperatures

Scalding vent fluid exceeds 400°C.

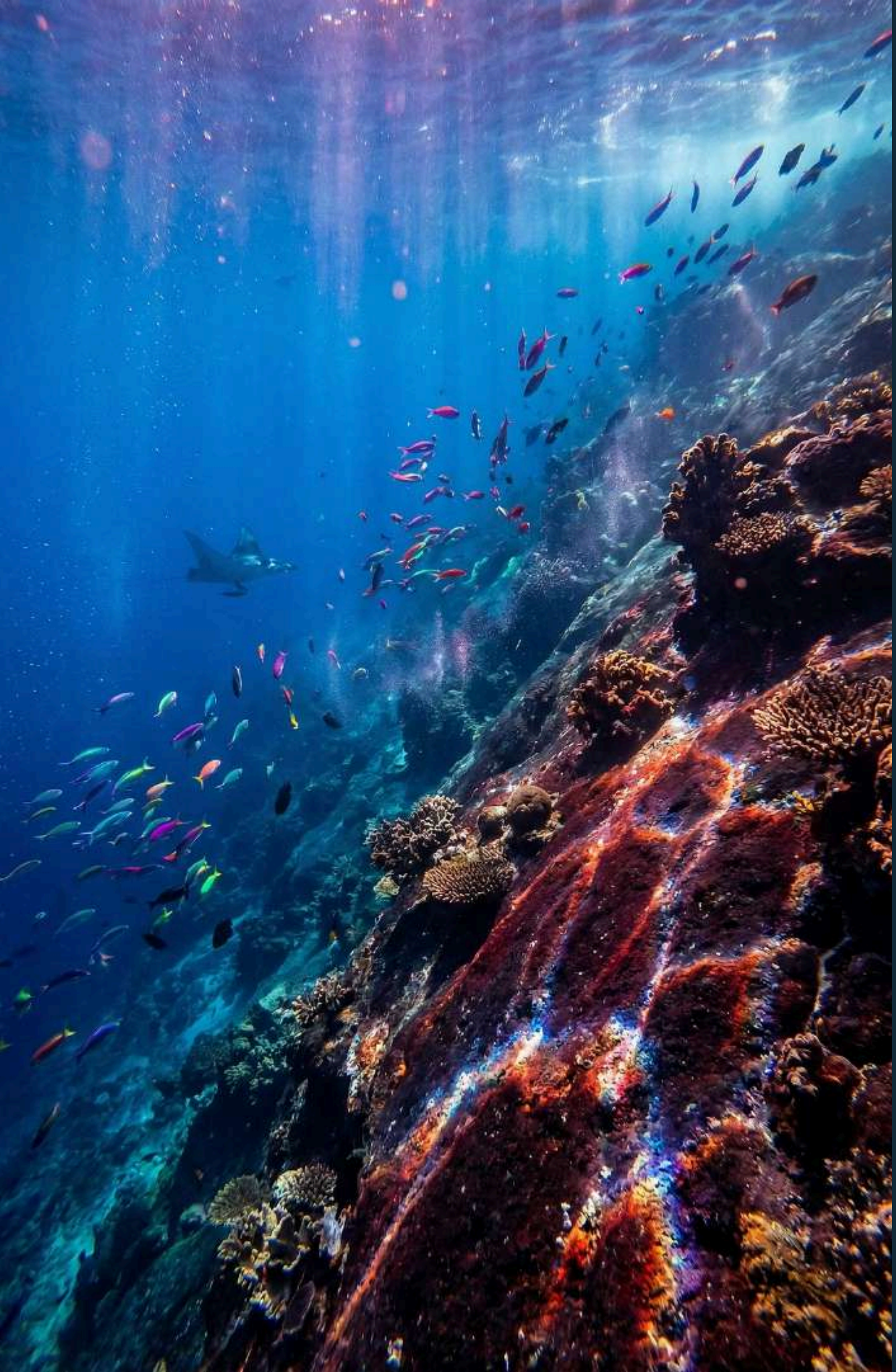
Toxic Chemicals & Acidic Conditions

High concentrations of hydrogen sulfide and heavy metals with low pH.

Crushing Pressure & Darkness

Extreme hydrostatic pressure and complete absence of sunlight.

Result: LOW BIODIVERSITY – Only specialized extremophiles, adapted to chemosynthesis, survive these intolerable conditions.



Example: Reef Slope

UNSTABLE & NON-EXTREME

The reef slope, or fore-reef, forms the seaward edge where coral reefs meet open ocean. Despite favorable temperatures and pH, physical instability limits biodiversity.

Limiting Factors

- Steep vertical walls face maximum wave and storm energy
- Sandy substrate easily eroded by strong currents and waves
- Limited primary producer biomass
- Short food chains with few trophic levels

Organisms present: Worms, clams, sand fleas, crabs

Result: LOW BIODIVERSITY

Ecosystem Comparison

Coral Reef	Stable	Non-extreme	**HIGH**
Rocky Shore	Stable	Non-extreme	**HIGH**
Sandy Shore	Unstable	Non-extreme	LOW
Hydrothermal Vent	Unstable	Extreme	LOW
Reef Slope	Unstable	Non-extreme	LOW

Notice the pattern: Stable + Non-extreme conditions consistently produce high biodiversity

🔔 Bell Ringer Check #1

Complete independently – 5 questions

1 [2 marks] Define biodiversity and state the three levels at which it can be measured.

2 [2 marks] Distinguish between species abundance and species richness.

3 [3 marks] Explain why reduced genetic diversity in a population could threaten its survival.

4 [3 marks] A coral reef is described as "stable and non-extreme." Explain what this means and why it leads to high biodiversity.

5 [2 marks] Why do rocky shores support greater biodiversity than sandy shores?

Service 2: Protection of Physical Environment

Marine ecosystems are natural barriers, safeguarding coastlines from the relentless forces of the ocean.

Seagrasses & Mangroves

- Root systems **stabilise** muddy substrate, preventing sediment loss.
- They **reduce erosion** of estuaries and mud flats, especially during storms.

Fringing Coral Reefs

- These reefs **absorb the power** of waves and storms before reaching the mainland.
- They **protect coastlines**, including sandy beaches, from weathering and erosion.

Service 3: Climate Control

Marine ecosystems regulate Earth's climate by absorbing significant CO_2, mitigating global warming.

Carbon Sequestration

Oceans are the largest carbon sink, absorbing ~25% of human-caused CO_2 emissions annually.

Phytoplankton Absorption

Phytoplankton absorb CO_2 via photosynthesis, vital for the global carbon cycle.

Blue Carbon Ecosystems

Mangroves, seagrasses, and salt marshes store "blue carbon" efficiently, reducing atmospheric CO_2.

Service 4: Providing Food Sources

Marine biodiversity is fundamental to the world's food supply, sustaining intricate ecosystems and directly feeding billions of people.

Primary Producers

Phytoplankton and macroalgae form the base of marine food chains, converting sunlight into energy.

Vibrant Food Webs

Complex interdependencies create stable ecosystems, efficiently moving energy through trophic levels.

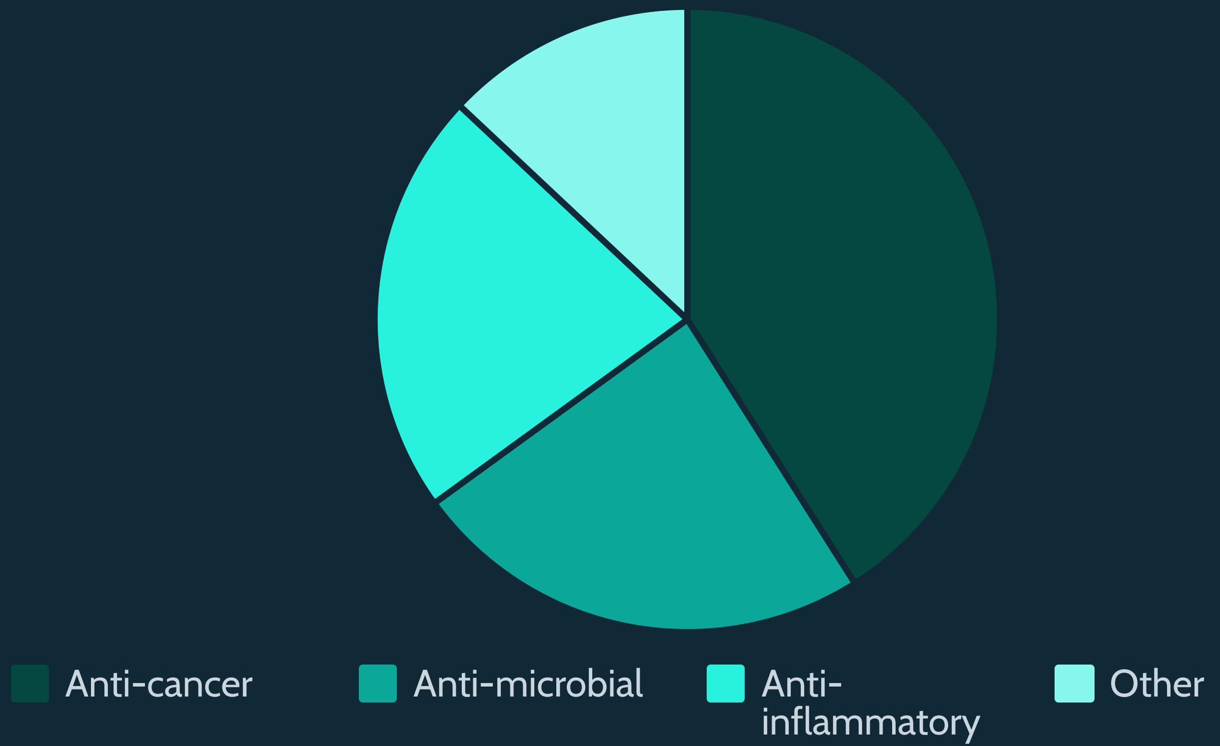
Human Food Sources

Humans depend heavily on the ocean's biological wealth, with marine species providing staple nutrition and economic sustenance worldwide.

- **Macroalgae:** (Seaweed) Used in cuisines and as industrial sources.
- **Crustaceans:** (Shrimp, Crab, Lobster) Globally popular shellfish.
- **Fish:** Critical protein and nutrient sources.

Service 5: Source of Medicines

The ocean functions as an "aquatic pharmacy," yielding compounds that treat diseases from cancer to infections. Marine biodiversity represents largely untapped medical potential.



Currently in Medical Trials

- Pain-relief drugs from marine cone snails
- Schizophrenia medications from marine worms
- Wound-healing compounds from corals
- Anti-cancer drugs from bacteria, bryozoans, fungi, tunicates, and nudibranchs

Source of Medicines

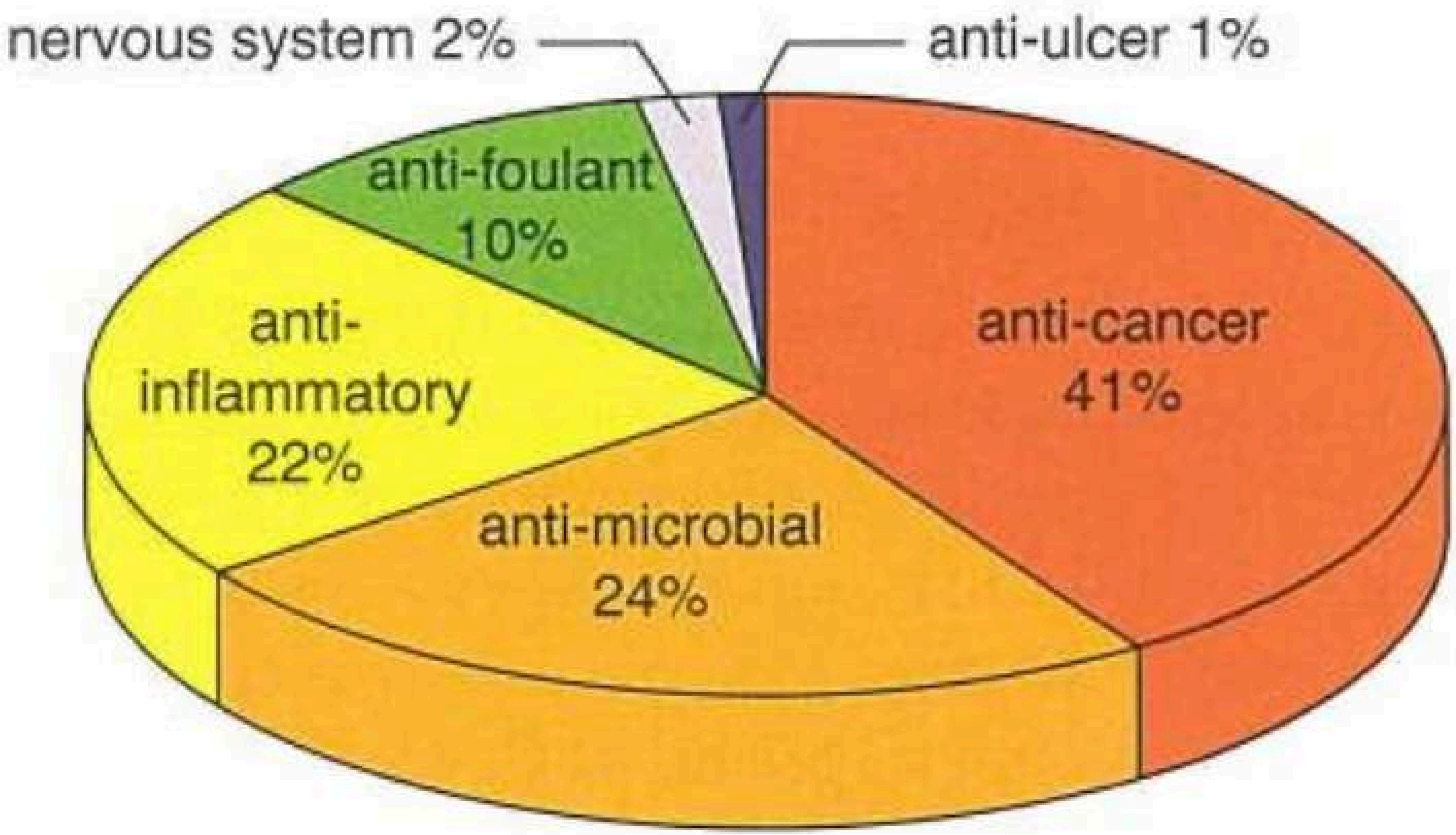


Figure 4.28: The use of beneficial chemicals derived from marine organisms.

Case Study: Keyhole Limpet Hemocyanin (KLH)

Source Organism

Giant keyhole limpet (*Megathura crenulata*), a large sea snail found along the Pacific coast

Active Compound

Keyhole limpet hemocyanin (KLH), a copper-containing protein that carries oxygen in the limpet's blood

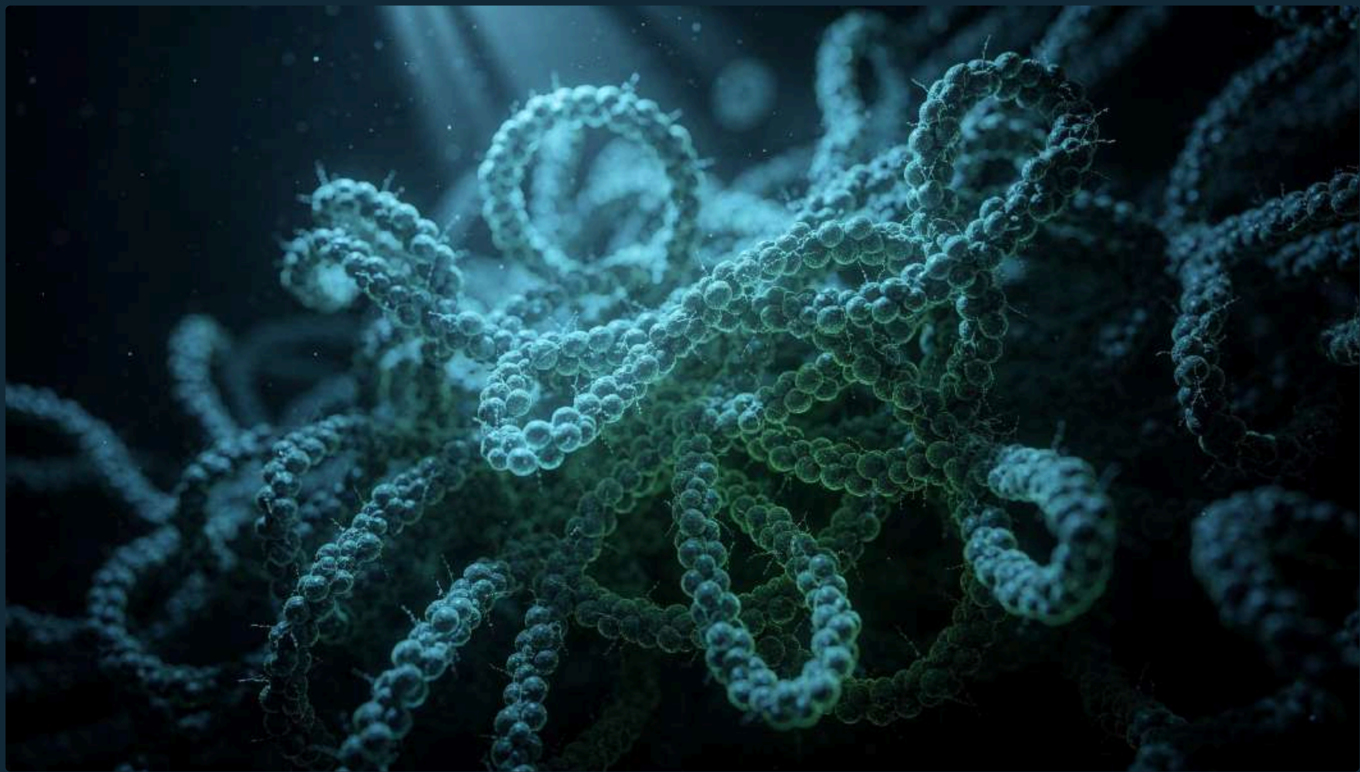
Medical Properties

Powerfully stimulates the human immune system, making it valuable for treating immune-related diseases

Current Applications

Used to treat autoimmune, inflammatory, and infectious diseases. Being tested as a vaccine for skin, breast, and bladder cancer.

Case Study: *Dermacoccus abyssi*



Source: The Deepest Place on Earth

A bacterium isolated from sediment in the Mariana Trench, the deepest part of the ocean at nearly 11,000 meters deep.

Discovery

This extremophile produces compounds called **dermacozines**, which show promise in fighting parasitic diseases.

Potential Application

May protect against the parasite causing African sleeping sickness (trypanosomiasis), a deadly disease affecting thousands annually.

Deep-sea trenches remain largely unexplored, representing vast potential for medical discoveries

Case Study: Australian Sea Sponges

Three species of Australian sea sponges produce a family of compounds called **chondropsins** that show remarkable medicinal potential across multiple diseases.

Bone Cancer

Inhibits enzymes involved in tumor growth

Alzheimer's Disease

Targets neural degeneration pathways

Viral Infections

Demonstrates antiviral properties

Diabetes

May help regulate blood sugar

Cardiovascular Disorders

Shows promise for heart disease treatment

Summary: Five Services of Marine Biodiversity

1. Stable Ecosystems	Coral reef food web complexity supports resilience
2. Physical Protection	Mangrove roots prevent coastal erosion during storms
3. Climate Control	Seagrass carbon storage mitigates climate change
4. Food Sources	Fish, crustaceans, and algae feed billions worldwide
5. Medicines	KLH from keyhole limpets for cancer treatment

Key Terms Review

Biodiversity: Measure of species, genetic, and ecosystem diversity

Species diversity: Abundance and richness of species in an area

Species abundance: Number of individuals per species

Species richness: Number of different species in an area

Genetic diversity: Variety of gene alleles within a species

Allele: Alternative form of a gene

Population: All individuals of one species in one area at one time

Allele frequency: How common an allele is in a population

Ecological diversity: Variation of ecosystems regionally or globally

Ecosystem: Community of organisms plus their abiotic environment

🔔 Bell Ringer Check #2

Complete independently – 5 questions

1 [3 marks] Describe how coral reefs and mangroves protect coastlines from damage.

2 [3 marks] Explain how phytoplankton and seagrasses help regulate climate change.

3 [2 marks] State two marine organisms that are sources of medicines and describe what they are used for.

4 [4 marks] Hydrothermal vents have low biodiversity but have provided potential medicines like dermacozines. Suggest why exploring extreme environments for new species is important.

5 [3 marks] A student claims "protecting marine biodiversity is only important for environmental reasons, not human health." Using named examples, argue against this claim.

What's Next?

COMING UP

4.4 Populations & Sampling Techniques

Populations & Communities

Defining and distinguishing these ecological units

Sampling Methods

Techniques for studying organisms in the littoral zone

Lincoln Index

Mark-recapture method for estimating population size

Data Analysis

Finding correlations and drawing conclusions